Dell PowerFlex v3.6.x

User Roles and LDAP Usage Technical Notes

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Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

PowerFlex LDAP

This publication presents a high-level description of the user roles and LDAP feature as implemented in PowerFlex and VxFlex Ready Node. Where features are relevant only for a specific system type, this is noted.

It is assumed that readers of this document are familiar with LDAP technology.

2

PowerFlex authentication and user roles

This section explains the user roles PowerFlex supports and how they are handled by the different authentication methods.

PowerFlex user roles

PowerFlex supports the following user roles:

- Superuser
- Security
- Administrator
- Configure (an aggregate of both FrontendConfigure and BackendConfigure) - native authentication only
- FrontendConfigure
- BackendConfigure
- Monitor

Each role is associated with different activities and commands, as explained in more detail in Command authorization. The manner in which roles are defined depends on whether native (local) or LDAP authentication is used, as explained in the following topic.

Local versus LDAP users

Local users function differently than LDAP users in PowerFlex. Although the role names are similar for local and LDAP users, the authorization permissions for each user role are defined differently for native and LDAP authentication.

Native authentication uses the nested model, in which a more privileged role immediately grants access to the same set of authorized commands associated with a less-privileged role, For example, in the local domain, when a user is defined as Administrator, that user may execute any command associated with the lesser privileged roles (Configure and Monitor). An Administrator may add users due to the Administrator role, and in addition, may also perform configuration changes through the Configure role.

For an LDAP user, this is not the case. Instead, a segregated model is used for the roles, and a specific set of commands is associated with each role. An LDAP user defined as an Administrator is limited to Administrator operations, such as adding users, but is not able to invoke any BackendConfigure, FrontendConfigure, or Monitor commands. You must assign LDAP users to each role to grant access to its specific set of commands.

The Configure and Superuser roles do not exist at all for LDAP authentication.

Local domain roles

When native authentication is used, all roles except Security are defined in a nested manner. Each role is authorized to perform operations related to its definition, in addition to the commands permitted by lower-level roles.

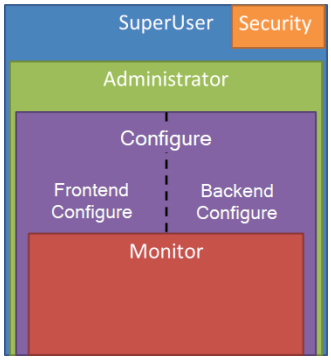


Figure 1. Local domain nested roles model

1. Superuser: Can perform all operations, configuration tasks, and upgrades. There can only be one Superuser, who is always a local user.
2. Security: Can define administrators and control LDAP. The Security user is not part of the hierarchical model and has a set of commands that only a Security (or Superuser) can perform.
3. Administrator: Can perform all Configure operations and also define the Configure- and Monitor-level users. Can perform upgrades.
4. Configure: Does not exist as a default option, but can be configured from FrontendConfigure or BackendConfigure. Can perform upgrades.
5. FrontendConfigure: Can perform Monitor operations and frontend-related configurations that include volume manipulations (adding, deleting, and mapping volumes and snapshots) and SDC operations (adding and deleting SDCs).
6. BackendConfigure: Can perform monitoring operations, upgrades, and backend-related configurations. Operations include:
   - Protection Domain and Storage Pool operations
   - Manipulating Fault Sets and SDS
   - Configuring SDS devices
   - Configuring replication
7. Monitor: Can only perform monitoring operations that do not affect the system.

NOTE: You must assign the BackendConfigure role to the users who need to update the system license.

LDAP domain roles

Roles in the LDAP domain are similar by name to local user roles, but are defined in a mutually exclusive manner.

Each role is authorized to perform a separate set of commands that are related to its functionality, with basically no overlap. The only exception is Configure, which is constructed from FrontendConfigure and BackendConfigure and does not exist as a separate role in LDAP. This model allows better granularity when defining users.

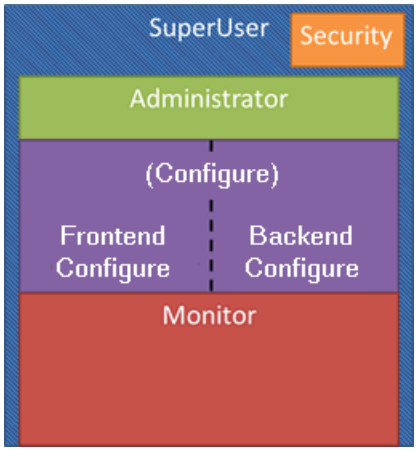


Figure 2. LDAP domain segregated roles model

1. Superuser: Not supported for LDAP users. Must be a local user.
2. Security: Can define administrators and control LDAP. The Security user is not part of the hierarchical model and has a set of commands that only a Security (or Superuser) user can perform.
3. Administrator: Can define Configure and Monitor users.
4. FrontendConfigure: Can perform any frontend-related configurations that include volume manipulations (adding, deleting, and mapping volumes and snapshots) and SDC operations (adding and deleting SDCs).
5. BackendConfigure: Can perform upgrades and any backend-related configurations. Operations include:
   - Protection Domain and Storage Pool operations
   - Manipulating Fault Sets and SDS
   - Configuring SDS devices
   - Configuring replication
6. Monitor: Can only perform monitoring operations that do not affect the system.

In order to run NDU, the LDAP user must be assigned at a minimum both the Monitor and BackendConfigure user roles.

NOTE: You must assign the BackendConfigure role to the users who need to update the system license.

LDAP login requirements

The following table lists the requirements for logging in to the different PowerFlex components using LDAP user roles:

Table 1. LDAP user roles by component

Component	Requirements
CLI, REST	At least one LDAP service in the system (not necessarily the one you log in with) must be assigned to the Administrator group. CLI and REST command execution will be according to the user role and permissions. Any LDAP user can log in (CLI command execution is according to the user role and permissions).
PowerFlex GUI, PowerFlex plug-in, PowerFlex Installer	Monitor role Additional roles can also be assigned in order to perform operations other than queries, but Monitor role is mandatory.
SRS	BackendConfigure role, Monitor role When performing system registration for SRS, the MDM remote syslog feature is enabled. BackendConfigure user role is required when accessing the MDM which will perform this operation. The MDM local user/LDAP user in the lockbox must be assigned Monitor privileges in addition to BackendConfigure privileges. The BackendConfigure role does not automatically include Monitor privileges.

Table 1. LDAP user roles by component (continued)

Component	Requirements
SNMP	BackendConfigure role, Monitor role The MDM local user/LDAP user in the lockbox must be assigned Monitor privileges in addition to BackendConfigure privileges. The BackendConfigure role does not automatically include Monitor privileges.

User permissions

The following table describes the permissions defined for each user role, depending on whether native or LDAP authentication is used:

Table 2. Permissions by user role

User role	Query		Configure parameters		Configure user credentials	
	Local	LDAP	Local	LDAP	Local	LDAP
Superuser (Only one Superuser is allowed per system, and it must be a local user.)	Yes	N/A	Yes	N/A	Yes	N/A
Security	No	No	No	No	Can assign Administrator users and control LDAP	
Administrator	Yes	No	Yes	No	Can assign Configure and Monitor users	
Configure (This role is only applicable for local users.)	Yes	N/A	Yes (an aggregate of FrontendConfigure and BackendConfigure)	N/A	No	N/A
FrontendConfigure	Yes	No	Yes Frontend operations only (volumes, SDCs, snapshots)		No	No
BackendConfigure	Yes	No	Yes Backend operations only (Protection Domains, Storage Pools, Fault Sets, SDSs, devices, other system settings)		No	No
Monitor	Yes	Yes	No	No	No	No

Superuser

The Superuser is the default user for setting up the system. The Superuser, which is only available in the local domain and cannot be an LDAP user, has all the privileges of all the roles.

Dell recommends using the Superuser role in the following cases:

- When setting up a new system.
- When there is no need to define separate or additional users.

The Superuser comes with the default username and password of admin/admin.

When a single Superuser is configured, the audit log does not track the activity of individual users logging in using the Superuser role.

Dell recommends disabling the Superuser after the system is deployed and other users are configured. This ensures that all users are associated with a specific person.

To disable the Superuser, from the SCLI run:

```
scli --disable_admin [--i_am_sure]
```

In emergency cases, you may need to override the LDAP authentication and restore the Superuser (admin) so that the authentication method becomes both LDAP and native authentication. To restore the Superuser, follow the "Reset the admin user password" procedure described in *Configure and Customize Dell PowerFlex* .

Changes to the LDAP server structure

When changes are made to the LDAP server structure after configuring LDAP in PowerFlex, you must reconfigure LDAP accordingly for the MDM, PowerFlex Gateway, and LIA, where applicable:

- Changes to the LDAP server's configuration, such as a new URI or base DN, require corresponding updates to the LDAP service, or adding a new LDAP service, on the MDM, PowerFlex Gateway, and LIA.
- Changes to LDAP groups require corresponding updates to the service groups on the PowerFlex clients.

Changes to user privileges take effect immediately in the PowerFlex client upon new login. However, users who are already logged retain their previous login privileges until the current session expires due to inactivity or exceeding the general session time limit.

3

Command authorization

This section lists the different user roles and gives examples of the commands associated with each role.

Administrator

The Administrator role is used to configure and monitor users. Examples of the commands associated with the Administrator user role are:

```
--add_user
--delete_user
--query_user
--query_users
--set_password
```

BackendConfigure

The BackendConfigure role allows users to perform operations such as configuring Protection Domains, Storage Pools, Fault Sets, SDRs, SDSs, and devices. Examples of commands associated with the BackendConfigure user role are:

```
--add_sds
--add_sds_device
--enable_rfcache
--enter_maintenance_mode
--inactivate_protection_domain
--modify_zero_padding_policy
--remove_protection_domain
--rename_storage_pool
--set_rebuild_policy
--add_sdr
--enter_sdr_maintenance_mode
--set_replication_journal_capacity
```

FrontendConfigure

The FrontendConfigure role allows users to perform frontend operations related to volumes, SDCs, and snapshots. Examples of commands associated with the FrontendConfigure user role are:

```
--add_sdc
--add_source_volume_to_snapshot_policy
--migrate_vtree
--modify_snapshot_policy
--overwrite_volume_content
--remove_volume
--resume_vtree_migration
--set_sdc_volume_limit
--snapshot_volume
--unlock_auto_snapshot
```

Configure

The Configure user role is available only in native authentication and is an aggregate of the FrontendConfigure and Backend Configure user roles.

Monitor

The Monitor role allows users to perform operations, such as queries, that do not affect the system. Examples of commands associated with the Monitor role are:

```
--query_all
--query_fault_set
--query_device_test
--query_diag_counters
--query_license
--query_network_latency_meters
--query_oscillating_failure_counter_parameters
--query_sds_network_test
--query_user_authentication_properties
--query_vtree_migration
```

Security

The Security role defines Administrator users and controls LDAP. Examples of commands associated with the Security role are:

```
--add_ldap_service
--assign_ldap_group_to_roles
--generate_certificate
--generate_mdm_certificate
--generate_mdm_csr_file
--remove_ldap_group_from_role_assignment
--remove_ldap_service
--replace_mdm_security_files
--set_cli_login_banner_acceptance_mode
--set_management_client_communication
```

4

Preparing PowerFlex nodes for use with LDAP

This section explains how to configure the servers on which PowerFlex components are installed (MDM or LIA) for use with LDAP. These configurations are required before configuring LDAP authentication on a PowerFlex node.

For instructions for preparing the PowerFlex Gateway for LDAP, see Configuring LDAP authentication on the PowerFlex Gateway.

Server configuration scenarios

PowerFlex supports LDAP authentication for three different purposes: LIA user authorization, MDM user authorization, and PowerFlex Gateway user authorization. Before configuring LDAP authentication for any of these purposes, perform the tasks in this section to prepare the PowerFlex servers or nodes that will use LDAP.

The following table explains the three LDAP cases and the stage in the PowerFlex deployment process at which you should prepare the server or nodes.

Table 3. LDAP use cases

LDAP feature	Description	Server/node	When to configure the server
LDAP for PowerFlex MDMs	PowerFlex uses LDAP to authorize users to configure, maintain, or monitor the system.	MDM nodes	Before or after system deployment, but before configuring LDAP authentication on the MDMs.
LDAP for PowerFlex Gateway	PowerFlex uses LDAP authentication for the PowerFlex Gateway.	PowerFlex Gateway	Before or after system deployment, but before configuring LDAP authentication on the gateway. **NOTE:** If you already prepared the gateway server for LDAP authentication for LIA, skip these tasks.
LDAP for PowerFlex LIA	LIA uses LDAP authorization to maintain and upgrade PowerFlex components.	PowerFlex Gateway	• New deployment: Prepare the server for LDAP before installing the PowerFlex Gateway. • Existing deployment: Prepare the server for LDAP before switching the LIA authentication mode from native to LDAP in the PowerFlex Installer. **NOTE:** If you already prepared the gateway server for LDAP authentication for the PowerFlex Gateway, skip these tasks.

LDAP server settings

Before preparing the server, ensure that you have the following information at hand for each LDAP server :

Table 4. Required LDAP server information

Setting	Configuration
IP address	
Username	
Password	
Domain name	
FQDN	
Certificate name	
Certificate location	

NOTE: PowerFlex supports both Linux- and Windows-based LDAPS servers. The Linux LDAP server can have multiple certificates.

Red Hat Enterprise Linux, CentOS, and SLES systems

Perform the following tasks in the order they are presented to set up a RHEL-, CentOS-, or SLES-based server for use with LDAP.

Prerequisites

Ensure the following:

- The OpenLDAP package is installed. (See the following instructions.)
- Each LDAP server has a separate base DN.

Steps

1. Run the following command to verify that the OpenLDAP package is installed.

   NOTE: OpenLDAP is a requirement for using LDAP on all Linux-based PowerFlex MDM nodes or for LIA, as stated in the *Getting to Know Dell PowerFlex Guide*

   ```
   rpm -qa | grep openldap
   ```

   Output similar to the following should appear, with the relevant package number:

   ```
   openldap-2.4.23-32.el6_4.1.x86_64
   ```

2. When secure LDAP is used, retrieve the configured CA certificate location from the OpenLDAP client configuration and make note of the location. For non-secure LDAP, skip to step 5.

   For example:

   ```
   grep TLS_CACERTDIR /etc/openldap/ldap.conf
   ```

   Output similar to the following should appear:

   ```
   TLS_CACERTDIR /etc/openldap/ldap.conf
   ```

   NOTE: The location of the CA certificates may differ depending on the Linux distribution and release.

3. If the certificate location does not exist, create a directory at the configured location: As root user, create a `cacerts` directory under `/etc/openldap`:

```
mkdir -p <TLS_CACERTDIR>
```

For example:

```
mkdir -p /etc/openldap/cacerts
```

4. Verify that the following entry is displayed in `/etc/openldap/ldap.conf`:
   - RHEL/CentOS:

     ```
     TLS_CACERTDIR /etc/openldap/cacerts/
     ```

   - SLES:

     ```
     TLS_CACERTDIR
     ```

5. On the PowerFlex component host, ensure that it is possible to establish a TCP connection to the LDAP server's address.
6. Import the certificate files from the LDAP server to the OpenLDAP client CA certificate location <TLS_CACERTDIR> determined in steps 2-4, and run `openssl rehash`. For non-secure LDAP, skip this task.

   For example:

   ```
   openssl rehash <TLS_CACERTDIR>
   ```

   For example:

   ```
   openssl rehash /etc/openldap/certs
   ```

   NOTE: The method of CA certificate import for the OpenLDAP client may differ based on your organization's security guidelines. Consult your security administrator for steps to import the CA certificates for use with LDAP client.

Install PowerFlex

Install the PowerFlex system.

For details, refer to the *Deploy Dell PowerFlex* guide.

Ubuntu

Perform the following tasks in the order they are presented to set up a Ubantu-based server for use with LDAP.

Configure LDAP nodes on the server

Use the following task to prepare Ubantu-based PowerFlex servera for use with LDAP.

Prerequisites

Ensure the following:

- The OpenLDAP package is installed.
- Each LDAP server has a separate base DN.

Steps

1. On the PowerFlex component host (MDM, Gateway or LIA), ensure that it is possible to establish a TCP connection to the LDAP server's address.

2. Run the following command, and wait until the operation is finished:

```
sudo apt-get update
```

3. Run:

```
sudo apt-get install openssh-client ldap-utils
```

4. In the OpenLDAP client configuration file, `/etc/ldap/ldap.conf`, find the value of the `TLS_CACERTDIR` parameter and write down the path. For non-secure LDAP, skip this step and jump to step 8.
5. Import the certificate files from the LDAP server to the certificate directory, `<TLS_CACERTDIR>`, configured in the LDAP client configuration file, `/etc/ldap/ldap.conf`.For non-secure LDAP, skip this step and jump to step 8.

   NOTE: The method of CA certificate import for the OpenLDAP client may differ based on your organization's security guidelines. Consult your security administrator for steps to import the CA certificates for use with LDAP client.

6. Run:

```
openssl rehash /etc/ssl/certs
```

   NOTE: For non-secure LDAP, skip this step and jump to step 8.

7. Run the following command and wait until the operation is finished:

```
sudo update-ca-certificates
```

   NOTE: For non-secure LDAP, skip to the next step.

8. Add the following lines to the `/etc/ldap/ldap.conf` file and ensure that all the other lines are commented out:

   `URI ldaps://example.ldaps.local/`

   or

   `URI ldap://server.example.com/`

   The file contents should be similar to the following example:

```
#
# LDAP Defaults
#

# See ldap.conf(5) for details
# This file should be world readable but not world writable.

#BASE    dc=example,dc=com
#URI    ldap://ldap.example.com ldap://ldap-master.example.com:666

#SIZELIMIT    12
#TIMELIMIT    15
#DEREF        never

# TLS certificates (needed for GnuTLS)
# TLS_CACERT    /etc/ssl/certs/ca-certificates.crt
URI ldaps://example.ldaps.local/
```

9. For each LDAP server's certificates, test that each certificate is valid.

```
sudo openssl s_client -connect example.ldaps.local:636 -CAfile /etc/ssl/certs/
scaleio.cer < /dev/null
```

   Output similar to the following appear. Confirm that the "`Verify return code`" line reads "`0 (ok)`".

```
CONNECTED(00000003)
depth=1 DC = local, DC = ldaps, CN = ldaps-example-CA
verify return:1
depth=0 CN = example.ldaps.local
```

```
verify return:1
. . .
---
New, TLSv1/SSLv3, Cipher is ECDHE-RSA-AES256-SHA384
Server public key is 2048 bit

Secure Renegotiation IS supported
Compression: NONE
Expansion: NONE
SSL-Session:
    Protocol  : TLSv1.2
    Cipher    : ECDHE-RSA-AES256-SHA384
    Session-ID: 000E0000A293DA7B903083F8BCB2737F2C10A98D298BD0E963B73E9EBAA4930C
    Session-ID-ctx:
    Master-Key:
2242522F8B151725BDED1DC92264F461F4487DAAEC8DB6F5EA4959A42D4DABD4105534110229870FC8B633
3B4443E891
    Key-Arg   : None
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    Start Time: 1455035763
    Timeout   : 300 (sec)
    Verify return code: 0 (ok)
---
DONE
```

Install PowerFlex

Install the PowerFlex system.

For details, refer to the *Deploy Dell PowerFlex* guide.

Windows

Perform the following tasks in the order they are presented to set up a Windows-based host server for using the LIA component with LDAP.

Confirm the LDAP server is accessible and ldp.exe is installed.

Ensure that the PowerFlex Gateway has access to the PowerFlex LDAP server and that `ldp.exe` is installed on the Gateway host.

Steps

1. On the PowerFlex Gateway host, ensure that it is possible to establish a TCP connection to the LDAP server's address.
2. Confirm that the `ldp.exe` tool is installed. If it is not present, install it:
   a. Open the Server Manager, and On the **Manage** menu of the Server Manager console, click **Add Roles and Features Wizard**, and then click **Next** three times until you see the **Add Roles and Features Wizard**.
   b. Select the **Active Directory Lightweight Directory Services** check box, and then click **Next**.

      NOTE: When you choose to install the Active Directory Lightweight Directory Services, it automatically selects and installs the .NET 3.5 Framework, unless it is already installed.

   c. Click **Add Features**, and then click **Next** three times, until the **Confirm installation selections** window appears.
   d. Click **Install**.
   e. When the installation finishes, click **Close**.

Import the CA certificates

If secure LDAP is used, import the CA certificate from the LDAP server into the local certificate store. When multiple LDAP servers are used, repeat this procedure to import the CA certificates for all the LDAP servers.

About this task

If non-secure LDAP is used, skip this task.

Steps

1. Copy the certificate file from the LDAP server to any location on the Windows server.
2. Double-click the certificate.

   The **Certificate** window appears:
3. Click **Install Certificate**, and then click **Next**.
   The **Certificate Import Wizard** window appears:

   NOTE: If an additional message appears, select **Local Machine**.
4. Select **Place all certificates in the following store**, and then click **Browse**.

   The **Select Certificate Store** window appears:
5. Select **Trusted Root Certification Authorities**, and then click **OK**.
6. Click **Next**.
7. Click **Finish**.
8. Click **OK** to confirm.

Test LDAP node connectivity using ldp.exe

Test and verify LDAP node connectivity using the `ldp.exe` utility.

Steps

1. From the command prompt run:

   ```
   ldp.exe
   ```

   The **Ldp** window appears:
2. From the **Connection** menu, select **Connect**.
3. In the **Connect** window, enter the following information:

   Secure LDAP:

   - **Server**: LDAPS server FQDN (for example, example.ldaps.local
   - **Port**: 636

   Non-secure LDAP:

   - **Server**: LDAP server FQDN (for example, example.ldap.local
   - **Port**:389
4. For secure LDAP, select the **SSL** check box, and then click **OK**. For non-secure LDAP, skip this step.
5. Confirm that the output displays connectivity to the LDAP server.

   If an error message appears, review the LDAP configuration.

Install PowerFlex

Install the PowerFlex system.

For details, refer to the *Deploy Dell PowerFlex* guide.

5

Configuring LDAP authentication on the PowerFlex MDM

This section contains instructions for configuring LDAP authentication on the PowerFlex MDM.

Preparing for LDAP authentication

Before configuring LDAP authentication on the PowerFlex MDM node, ensure that the system meets the following prerequisites:

- You have prepared both the LDAP server and the server on which the MDM is to be installed. For instructions, see Preparing PowerFlex nodes for use with LDAP.
- PowerFlex has been deployed on the node.

Configuration overview

This topic provides a high-level overview of the CLI commands and special considerations to take into account when configuring LDAP authentication on the MDM. For the specific command syntax, including command parameters, for your LDAP configuration, refer to the relevant tasks immediately following this topic.

Add the LDAP server as a service

Add the LDAP server as a service using the `--add_ldap_service` command.

Assign the LDAP groups to roles

After the LDAP service is configured, use the `--assign_ldap_groups_to_roles` command to map the LDAP groups to PowerFlex user roles. Assign an LDAP group to each PowerFlex role. The same LDAP group can be assigned to multiple roles. For an explanation of the different user roles, see PowerFlex authentication and user roles.

The following considerations should be made when assigning groups to roles when LDAP authentication is used:

- If you want LDAP users to be able to use the PowerFlex GUI or query the system, you must assign an LDAP group to the Monitor role.
- An LDAP user defined as an Administrator is limited to Administrator operations, such as adding users, but cannot invoke any Configure commands. To overcome this limitation, assign an LDAP group to both the Administrator and Configure roles.
- If the LDAP user will run NDU (non-disruptive upgrade), assign the user at a minimum to both the Monitor and BackendConfigure user roles.

NOTE: If an LDAP user role is changed, users must log out of PowerFlex and log back in with the updated permissions.

Set the user authentication method

After the LDAP service is set and groups are assigned, use the `--set_user_authentication_method` command to specify the authentication method with which PowerFlex will authenticate the users.

You may restrict users to only the local domain (native authentication) or to LDAP authentication only, or you can configure the MDM to allow both types of users. This decision should be made at the discretion of the system administrator, and is usually dictated by the security policy of the organization.

NOTE: After the authentication method is set exclusively to LDAP, it cannot be changed easily back to native authentication. In cases where access to the LDAP server is not possible and the authentication method must be changed back to native, see the "Reset the admin user password" procedure described in *Configure and Customize Dell PowerFlex*

Log in to the system

After configuring the system parameters, you may log in to the system using the `scli --login` command.

When logging in as a local user, the command must include a user name. When using LDAP, the command should also include the relevant LDAP domain and the LDAP authentication parameter.

Example for local login using native authentication:

```
scli --login --username JohnDoe
```

Example for LDAP login:

```
scli --login --username JohnDoe@example.ldaps.local --ldap_authentication
```

Verify the configuration

The `--query_user_authentication_properties` command returns the configuration so you can confirm that the configuration is correct.

Configure LDAP authentication on an MDM node

Use the following procedure to configure LDAP authentication using the PowerFlex CLI (SCLI). The actual command syntax should match your system's environment.

About this task

For instructions on accessing the CLI and a detailed explanation of the commands and their parameters, see CLI command details.

Steps

1. Add the LDAP service to the MDM:

```
scli --add_ldap_service --ldap_service_uri <URI> --ldap_base_dn <LDAP_DN> [--
ldap_fqdn <FQDN>] [--ldap_service_name <LDAP_NAME>] [--username_dn_format <FORMAT>]
[--search_filter_format <FORMAT>] [--authorizer_dn <USER_DN>] [--authorizer_password
<PASSWORD>]
```

NOTE: Configuring multiple LDAP servers with the same base DN is not supported.

2. Assign the LDAP groups to roles:

```
scli --assign_ldap_groups_to_roles (--ldap_service_id <LDAP_SERVICE_ID>
| --ldap_service_name <LDAP_SERVICE_NAME>)  [--administrator_role_dn
<LDAP_GROUP_DN>] [--security_role_dn <LDAP_GROUP_DN>] [--backend_config_role_dn
[--frontend_config_role_dn <LDAP_GROUP_DN>] [--monitor_role_dn <LDAP_GROUP_DN>]
<LDAP_GROUP_DN>] [--allow_overwrite ]
```

NOTE: You can assign only one LDAP group to each role.

3. Set the user authentication method:

```
scli --set_user_authentication_method (--ldap_authentication | --
native_authentication | --native_and_ldap_authentication | allow_ldap_without_admin)
[--i_am_sure]
```

4. Verify the configuration:

```
 scli --query_user_authentication_properties
```

Ensure that the output matches your system environment and no errors are displayed. For example:

```
------------------------------------------------------------
User authentication method: Native and LDAP
System has 1 configured LDAP services
------------------------------------------------------------
LDAP service ID: 226521b800000000
LDAP service name: linux_ldap
LDAP service URI: ldaps://server.ldap.com
Users base DN: dc=ldap,dc=com
FQDN: ldap.com
User search filter: (&(objectClass=userSecurityInformation)(cn=[USER])
(memberOf=[GROUP]))
Username DN format: CN=[USER],OU=People,DC=ldap,DC=com
LDAP service has 1 configured groups.
        Role: Administrator
        Group DN: cn=scaleio_QA,ou=People,dc=ldap,dc=com
------------------------------------------------------------
```

Cross domain authentication

When there are two or more LDAP servers with a trusted relationship, a user from one server (user_ldap_server) can be associated with a group (powerflex_group) that is configured in the second server (group_ldap_server). The group in the second server is associated with a role in the PowerFlex system.

About this task

For PowerFlex to allow this user to log in, perform the following steps:

Steps

1. Add the user_ldap_server as an LDAP service in PowerFlex.
2. Assign the powerflex_group (that belongs to the group_ldap_server) to the PowerFlex LDAP service of user_ldap_server, and associate it with the desired role.

Results

From now on, any user from the user_ldap_server can log in to PowerFlex with the role associated with powerflex_group that belongs to group_ldap_server.

Example of cross domain authentication

This example demonstrates the configuration performed in PowerFlex.

About this task

The example assumes that a user is a defined user in user_ldap_server. In addition, a group named powerflex_monitor_group is defined in group_ldap_server.

The following steps show how to enable access to PowerFlex systems for users in the user_ldap_server (e.g., John Doe) who are part of the powerflex_monitor_group defined in group_ldap_server.

Steps

1. Add user_ldap_server as an LDAP service.

   For example:

   ```
   scli --add_ldap_service --ldap_service_uri "ldaps://server.ldap.com"
   --ldap_base_dn "dc=ldap,DC=com" --ldap_service_name "example_ldap" --
   username_dn_format "CN=[USER],OU=People,DC=ldap,DC=com" --search_filter_format
   "(&(objectClass=userSecurityInformation)(cn=[USER])(memberOf=[GROUP]))"
   ```

   NOTE: Configuring multiple LDAP servers with the same base DN is not supported.

   Output similar to the following should appear:

   ```
   Successfully added an LDAP service.  Object ID fe97b19a00000000 with Name:
   linux_ldap, URI: ldaps://server.ldap.com and base DN: dc=ldap,DC=com
   ```

2. Assign the powerflex_monitor_group group to the Monitor role:

   ```
   scli --assign_ldap_groups_to_roles --
   ldap_service_name "example_ldap" --monitor_role_dn
   "CN=powerflex_monitor_group,OU=powerflex_ou,DC=group_ldap_server,DC=example,DC=com"
   ```

   The following output should appear:

   ```
   Assignment of roles to LDAP groups completed successfully
   ```

3. Check the settings:

   ```
   scli --query_user_authentication_properties
   ```

   Output similar to the following should appear:

   ```
   ------------------------------------------------------------
   System has 1 configured LDAP services
   ------------------------------------------------------------
   LDAP service ID: fe97b19a00000000
   LDAP service name: example_ldap
   LDAP service URI: ldaps://server.ldap.com
   Users base DN: dc=ldap,DC=com
   LDAP service has 1 configured groups.
              Role: Monitor
   Group DN:
   CN=powerflex_monitor_group,OU=powerflex_ou,DC=group_ldap_server,DC=example,DC=com
   ------------------------------------------------------------
   ```

4. Set the authentication method:

   ```
   scli --set_user_authentication_method --native_and_ldap_authentication
   ```

   The following output should appear:

   ```
   Authentication method changed successfully
   ```

5. Log in using the desired user name and the assigned authentication mode for the user:

   ```
   scli --login --username "JohnDoe@example.ldaps.local" --password <password> --
   ldap_authentication
   ```

   Output similar to the following should appear:

   ```
   Logged in. User role is Monitor. System ID is 31bf07056dd2f5d7
   ```

CLI command details

This section explains how to access and log in to the CLI. It also provides detailed information regarding the commands used to configure LDAP authentication on the PowerFlex MDM.

Using the CLI

The PowerFlex CLI (SCLI) is installed as part of the MDM component and can be found in the following path: `scli`

All CLI commands use the following format:

```
scli [--mdm_ip <IP>] <command>
```

For more information, see *PowerFlex CLI Reference Guide*.

add_ldap_service

Add the service to the MDM to be used for authentication. In return, the user receives the ID of the LDAP service.

NOTE: LDAP should be configured on all the MDMs in the system in order to support switch ownership scenarios. For more information, see Preparing PowerFlex nodes for use with LDAP.

NOTE: PowerFlex systems support authentication by up to eight LDAP servers. When multiple LDAP servers are used, add each one separately using this command.

Syntax

```
scli --add_ldap_service --ldap_service_uri <URI> --ldap_base_dn <LDAP_DN> [--ldap_fqdn
<FQDN>]
  [--ldap_service_name <LDAP_NAME>]
  [--username_dn_format <FORMAT>]
  [--search_filter_format <FORMAT>]
  [--authorizer_dn <USER_DN>]
  [--authorizer_password <PASSWORD>
```

NOTE: `--ldap_fqdn`, `--username_dn_format`, and `--search_filter_format` parameters are used when the LDAP server is running on Linux.

Parameters

`--ldap_service_uri <URI>`

URI of the LDAP service:

<LDAP_SCHEMA>://<LDAP_HOSTNAME>: [*<PORT_NUMBER>*]

Where:

<LDAP_SCHEMA>

Defines the connection protocol:

- LDAPS: Secure LDAP connection (recommended)
- LDAP: non-secure LDAP connection

<LDAP_HOSTNAME>

LDAP hostname

<PORT_NUMBER>

LDAP service port (optional, default: 389 for LDAP and 636 for LDAPS)

This parameter must start with ldap:// or ldaps:// followed by the host name.

Example: ldaps://my.ldaphost.com:636

When using the global catalog, one should specify the port number of the global catalog. The standard ports for global catalog are 3268 (non-secure) and 3269 (secure).

NOTE: No extra validation is performed at this stage.

--ldap_base_dn *<LDAP_DN>*

Base Distinguished Name (DN) of users in the domain. Must be a valid DN containing the DC substring. For example, if a user corporate login is johnd@ecme.corp.com, the DC string would be DC=ecme, DC=corp, DC=com.

NOTE: Configuring multiple LDAP servers with the same base DN is not supported.

NOTE: On Active Directory Windows servers, use the dsquery tool to find LDAP Base DN information. To see available options, in the command line type `dsquery /?`

On Linux servers, from the command line, use `ldapsearch`. (ldapsearch may need to be installed.)

--ldap_fqdn *<FQDN>*

The FQDN is used to identify the LDAP service. By default it is derived from the base-DN, but there are cases that it must be defined explicitly.

If you want to support multiple FQDNs for one service, this parameter can contain a list of sub-strings in square brackets. For example: `[us.,eu.,as.,]dell.ldap` will include users with the following suffixes: us.dell.ldap, eu.dell.ldap, as.dell.ldap, dell.ldap.

--ldap_service_name *<LDAP_NAME>*

LDAP service name

--username_dn_format *<FORMAT>*

The username format in DN format (must contain [USER] as a place holder for the username)

--search_filter_format *<FORMAT>*

A search filter for the LDAP query. This is required only if it is different from the default. It must contain [USER] and [GROUP] as place holders for username and group-DN. Example: `(&(objectClass=user)(sAMAccountName=[USER])(memberOf:1.2.840.113556.1.4.1941:=[GROUP]))`

--authorizer_dn *<USER_DN>*

The authorizer username for groups search

--authorizer_password *<PASSWORD>*

Password of the authorizer user

Examples

```
scli --add_ldap_service --ldap_service_uri "ldaps://ldaps.ecme.com:636" --ldap_base_dn
"OU=SIO_OU_1,DC=ldaps,DC=local"
```

where:

- `ldaps://ldaps.ecme.com` is the host name of the authentication server.
- 636 is the port number.
- `OU=SIO_OU_1` is a specific organizational unit group defined in the Active Directory.
- `DC=ldaps, DC=local` are the domain component parts of the Base DN.

```
scli --add_ldap_service --ldap_service_uri ldaps://ldaps.ecme.com:3269 --ldap_base_dn
ou=sio_ou_1,dc=ldaps,dc=ecme,dc=com --ldap_fqdn [na.,eu.,as.,]ldaps.ecme.com
```

where:

- The access will be to the global catalog through port 3269.
- `ou=sio_ou_1` is a specific organizational unit group defined in the Active Directory.
- `dc=ldaps,dc=ecme,dc=com` are the domain component parts of the Base DN.
- The following FQDNs servers will be mapped to the same URI ldaps://ldaps.ecme.com(): ldaps://na.ldaps.ecme.com, ldaps://eu.ldaps.ecme.com, ldaps://as.ldaps.ecme.com, and ldaps://ldaps.ecme.com.

assign_ldap_groups_to_roles

Map LDAP groups to PowerFlex system roles.

The LDAP service must be configured before using this command. Once you have mapped the roles, you can assign users in the Active Directory to the relevant LDAP groups.

NOTE: To enable LDAP users to use the PowerFlex GUI or vSphere Plug-in, you must assign all LDAP groups the Monitor role.

Syntax

```
scli --assign_ldap_groups_to_roles (--ldap_service_id <LDAP_SERVICE_ID> | --
ldap_service_name <LDAP_SERVICE_NAME>)
  [--administrator_role_dn <LDAP_GROUP_DN>]
  [--security_role_dn <LDAP_GROUP_DN>]
  [--backend_config_role_dn <LDAP_GROUP_DN>]
  [--frontend_config_role_dn <LDAP_GROUP_DN>]
  [--monitor_role_dn <LDAP_GROUP_DN>]
  [--allow_overwrite ]
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

--administrator_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with administration privileges

--security_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with security privileges

--backend_config_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with backend configuration privileges

--frontend_config_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with frontend configuration privileges

--monitor_role_dn *<LDAP_GROUP_DN>*

LDAP group that has users with monitoring privileges

--allow_overwrite

Overwrites the role's LDAP group

Example

```
scli --assign_ldap_groups_to_roles --ldap_service_id 0xAABBCCDDEEFF0011
--administrator_role_dn "CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com" --
monitor_role_dn "CN=SIO_GRP_2,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com"
```

set_user_authentication_method

Set the user authentication method for the system.

CAUTION: Use this command with caution. The operation is complex to roll back.

Syntax

```
scli --set_user_authentication_method (--ldap_authentication | --native_authentication |
--native_and_ldap_authentication | allow_ldap_without_admin)
  [--i_am_sure]
```

Parameters

--ldap_authentication

LDAP-based authentication method where users are managed on an LDAP-compliant server. Configure LDAP service and LDAP user before switching to this authentication method.

--native_authentication

Native authentication method where users are managed locally in the system

--native_and_ldap_authentication

A hybrid authentication method. Both LDAP and local users may log in to the system after it is set.

--allow_ldap_without_admin

Allow setting LDAP athentication method even if there is no LDAP service with administrator role.

--i_am_sure

Skip the safety questions for command execution. (For example: “This could damage the stored data. Are you sure?”)

Example

```
scli --set_user_authentication_method --native_and_ldap_authentication --i_am_sure
```

query_user_authentication_properties

Retrieve information about LDAP services configured in the system.

Syntax

```
scli --query_user_authentication_properties
   --mdm_port 6611
```

```
scli --query_user_authentication_properties
```

Parameters

None.

Example

```
scli --query_user_authentication_properties --mdm_port 6611
```

```
scli --query_user_authentication_properties
```

Additional PowerFlex CLI commands for LDAP

This section contains additional PowerFlex CLI commands used to maintain and modify LDAP authorization on the MDM.

remove_ldap_group_from_role_assignment

Remove LDAP groups from LDAP user role assignments.

Syntax

```
scli --remove_ldap_group_from_role_assignment (--ldap_service_id <LDAP_SERVICE_ID> | --
ldap_service_name  <LDAP_SERVICE_NAME>)
  [--administrator_role]
  [--security_role]
  [--backend_config_role]
  [--frontend_config_role]
  [--monitor_role]
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

Options:

(choose at least one of the following)

--administrator_role

LDAP group containing users with administration privileges

--security_role

LDAP group containing users with security privileges

--backend_config_role

LDAP group that containing with backend configuration privileges

--frontend_config_role

LDAP group that containing with frontend configuration privileges

--monitor_role

LDAP group that containing with monitoring privileges

Example

```
scli --remove_ldap_group_from_role_assignment --ldap_service_id 0xAABBCCDDEEFF0011 --
administrator_role "CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=ecme,DC=com"
```

rename_ldap_service

Assign a name to, or rename, an LDAP service.

Syntax

```
scli --rename_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME>) --new_name <NAME>
   --mdm_port 6611
```

```
scli --rename_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME>) --new_name <NAME>
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of LDAP service

--new_name *<NAME>*

New name to be assigned to the LDAP service

Example

```
scli --rename_ldap_service --ldap_service_name prevLSName --new_name newLSName --
mdm_port 6611
```

```
scli --rename_ldap_service --ldap_service_name prevLSName --new_name newLSName
```

remove_ldap_service

Remove an LDAP service from the system.

Syntax

```
scli --remove_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME> | --remove_all)
   --mdm_port 6611
```

```
scli --remove_ldap_service (--ldap_service_id <LDAP_SERVICE_ID> | --ldap_service_name
<LDAP_SERVICE_NAME> | --remove_all)
```

Parameters

--ldap_service_id *<LDAP_SERVICE_ID>*

ID of the LDAP service

--ldap_service_name *<LDAP_SERVICE_NAME>*

Name of the LDAP service

--remove_all

Remove all LDAP services.

Example

```
scli --remove_ldap_service --ldap_service_name ldap1 --mdm_port 6611
```

```
scli --remove_ldap_service --ldap_service_name ldap1
```

6

Configuring LDAP authentication on the PowerFlex Gateway

This section explains how to configure LDAP authentication for the PowerFlex Gateway/PowerFlex Installer.

Preliminary configurations

Before configuring LDAP authentication on the PowerFlex Gateway, ensure that the system meets the following prerequisites:

- You have prepared both the LDAP server and the server on which the PowerFlex package is to be installed. For instructions, see Preparing PowerFlex nodes for use with LDAP.
- The PowerFlex Gateway package has been deployed on the server.

FOSGWTool basics

FOSGWTool is a gateway-related command line script that enables you to run PowerFlex Gateway-commands.

To use FOSGWTool, enter the appropriate path based on the gateway's operating system, and append the commands to the end of the path. FOSGWTool is located at:

- Linux:

```
/opt/emc/scaleio/gateway/bin/FOSGWTool.sh
```

- Windows:

```
C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat
```

Establish trust between the PowerFlex Gateway and the LDAP server

If secure LDAP is used, configure a secured connection to the LDAP servers by adding the LDAPS servers' certificates to the PowerFlex Gateway's truststore using the Java Keytool utility.

About this task

Keytool is a part of the Java (JRE or JDK) installation and can be found in the `bin` directory. For more information about using Keytool, default paths, and typical commands, see "Using Keytool to add certificates to external components" in the *Configure and Customize Dell PowerFlex* .

Steps

1. Add the LDAPS servers' certificates to the PowerFlex Gateway's `truststore.jks` file.

```
keytool -importcert -trustcacerts -alias [unique_alias] -file
[path_to_the_certificate_file] -keystore [path_to_certificates_folder]/truststore.jks
```

The `truststore.jks` is located at:

- Linux: `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/certificates/truststore.jks`
- Windows: `C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\certificates\truststore.jks`

For multiple LDAP servers, import all the certificates. Each certificate alias must be unique. Dell recommends using the certificate's full subject. For example:

`givenname=mdm, ou=asd, o=emc, l=hopkinton, st=massachusetts, c=us, cn=centos-6.4-adi5`

The default truststore password is `changeit`.

2. Restart the PowerFlex Gateway service:
   - On a Linux server run:

     ```
     service scaleio-gateway restart
     ```

   - On a Windows server: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
3. Run the following command and confirm in the output that each LDAPS server's certificate was imported successfully into the `truststore.jks` file:

   ```
   keytool -list -v -keystore [truststore.jks_path] -storepass <storepass>
   ```

   NOTE: The default storepass is `changeit`.

   For example:

   ```
   keytool -list -v -keystore "/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/
   certificates/truststore.jks" -storepass changeit
   ```

Configure LDAP authorization on the PowerFlex Gateway

Use the following task to prepare the PowerFlex Gateway and configure LDAP authorization on a PowerFlex Gateway.

Prerequisites

Ensure the following:

- The PowerFlex Gateway host can establish a TCP connection to the LDAP server's address.
- Each LDAP server has a separate base DN.

About this task

You can configure up to eight LDAP servers. When adding multiple LDAP servers, ensure that the list for each parameter in the `reset_ldap_properties` command is the same length. Items in lists should be separated by "`;`". When multiple LDAP servers are used, you must import each server's certificate into the gateway's truststore, as described in the previous task.

NOTE: The `FOSGWTool --set_ldap_properties` command that existed in previous PowerFlex versions has been deprecated, and is now supported for only a single LDAP server. You cannot use it to add multiple LDAP servers.

Steps

1. In FOSGWTools, run the `--reset_ldap_properties` command to add the LDAP parameters to the PowerFlex Gateway.

   Syntax:

   ```
   FOSGWTool.sh --reset_ldap_properties [--server_url_list <URL_LIST>] [--base_dn_list
   <BASE_DN_LIST>] [--group_name_list <GROUP_NAME_LIST>] [--dn_format_list
   <DN_FORMAT_LIST>] [--filter_list <FILTER_LIST>] [--authorizer_dn_list
   <AUTHORIZER_DN_LIST>] [--authorizer_password_list <AUTHORIZER_PASSWORD_LIST>] [--
   create_default_lockbox]
   ```

   If a previous LDAP configuration exists, an error message is returned. To confirm overwriting of the configuration, issue the command again, with the flag `--i_am_sure` appended to the command.
2. Optionally view the existing LDAP configuration:

- Linux gateway:

```
/opt/emc/scaleio/gateway/bin/FOSGWTool.sh --query_ldap_properties
```

- Windows gateway:

```
C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat --query_ldap_properties
```

Disable local authentication on the PowerFlex Gateway

If only LDAP authentication will be used, and native authentication will be disabled, use the following task to disable local authentication in the `gatewayUser.properties` file. If local authentication should remain available, skip this task.

Steps

1. Using a text editor, open the `gatewayUser.properties` file.

   The `gatewayUser.properties` file is located in the following directory on the PowerFlex Gateway server:

   - Linux: `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes`
   - Windows: `C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\`
2. Change the `gateway-admin.disable.local.login` property to "true":

   `gateway-admin.disable.local.login=true`.
3. Save and close the file.
4. Restart the PowerFlex Gateway service:

   - On a Linux server run:

     ```
     service scaleio-gateway restart
     ```

   - On a Windows server: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.

Results

When the configuration is complete, you must log in to the PowerFlex Gateway using LDAP user credentials. For example:

```
user: sio_1@ldaps.local, password: password
```

FOSGWTool commands

The following commands are used to configure LDAP authentication for the PowerFlex Gateway.

reset_ldap_properties

Save multiple LDAP servers in a lockbox. If explicity noted, create a lockbox first.

You can save the properties for up to eight LDAP servers. When adding multiple LDAP servers, ensure that the list for each parameter in the `reset_ldap_properties` command contains the same number of entries. Separate between entries in each list with an `";"`.

Syntax

```
FOSGWTool.sh  --reset_ldap_properties [--server_url_list <URL_LIST> --base_dn_list
<BASE_DN_LIST> --group_name_list <GROUP_NAME_LIST> [--dn_format_list <DN_FORMAT_LIST>]
[--filter_list <FILTER_LIST>] [--authorizer_dn_list <AUTHORIZER_DN_LIST>] [--
authorizer_password_list <AUTHORIZER_PASSWORD_LIST>] [--create_default_lockbox]
```

NOTE: The FOSGWTool for Windows is `FOSGWTool.bat`.

Parameters

`--server_url_list<URL>`

(Mandatory.) List of LDAP server URLs:

<ldap-schema>://<ldap-hostname>: <port-number>

Where:

<ldap-schema>

Defines the connection protocol:

- LDAPS for secure connection (recommended) , or
- LDAP for non-secured TCP connection

<ldap-hostname>

LDAP hostname

<port-number>

LDAP service port (optional)

Example: ldaps://win12r2-dc.ldaps.local;ldap://win08r2-dc.ldap.local

`--base_dn_list <BASE_DN>`

(Mandatory.) List of base Distinguished Names (DN) of users in domains.

NOTE: On Active Directory Windows servers, use the dsquery tool to find LDAP Base DN information. To see available options, in the command line type `dsquery /?`

On Linux servers, from the command line, use `ldapsearch`. (Ldapsearch may need to be installed.)

`--group_name_list <GROUP_NAME>`

(Mandatory.) List of LDAP groups that contain users with administration privileges

`--dn_format_list <FORMAT>`

(Optional.) List of DN (Distinguished Names) formats. Relevant only for Linux servers. If not defined, default is used.

`--filter_list <FILTER>`

(Optional.) List of filters (Distinguished Names). Relevant only for Linux servers. If not defined, default is used.

`--authorizer_dn_list <USER_DN>`

(Optional.) List of user DNs that are authorized to perform an LDAP server search

`--authorizer_password_list <PASSWORD>`

(Mandatory if `--authorizer_dn_list` is define.) List of passwords corresponding to the list of Authorizer DNs. Because list items are separated with a semi-colon (;), passwords cannot contain the ";" character.

`--i_am_sure`

Gives preemptive approval to the command

`--create_default_lockbox`

Creates a lockbox with a random passphrase if one doesn't already exist

Single LDAP server example

```
FOSGWTool.sh --reset_ldap_properties --server_url_list "ldap://server.ldap.com" --
base_dn_list "dc=ldap,dc=com" --group_name_list "cn=scaleio_QA,ou=People,dc=ldap,dc=com"
--dn_format_list "CN=<USER>,OU=People,DC=ldap,DC=com" --filter_list "(&(objectClass=*)
(cn=<USER>)(memberOf=<GROUP>))
```

Multiple LDAPS server examples

```
FOSGWTool.sh  --reset_ldap_properties --i_am_sure --
server_url_list "ldap://win12r2-dc.ldaps.local;ldap://server.ldap.com" --
base_dn_list "DC=ldaps,DC=local;dc=ldap,dc=com" --group_name_list
"CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=local;cn=scaleio_QA,ou=People,dc=ldap,dc=com" --
dn_format_list ";CN=<USER>,OU=People,DC=ldap,DC=com" --filter_list ";(&(objectClass=*)
(cn=<USER>)(memberOf=<GROUP>))
```

When adding multiple LDAP servers, ensure that the list for each parameter in the `--reset_ldap_properties` command is the same length. Items in lists should be separated with a ";".

NOTE: In the above example for multiple servers, the first server in the list is a Windows LDAP server, whereas the second server in the list is a Linux LDAP server. Because the `--dn_format_list` and `--filter_list` parameters are relevant only to Linux LDAP servers and not for Windows LDAP servers, an empty field followed by the ";" is used for the Windows server's value in each of these parameter lists..

update_ldap_properties

Check whether the LDAP properties in the lockbox are configured in the format for earlier versions of PowerFlex. If an LDAP property, such as the group name, is detected in the old format, the command updates the property to the currently supported format.

Syntax

```
FOSGWTool.sh -update_ldap_properties
```

NOTE: The FOSGWTool for Windows is `FOSGWTool.bat`.

Example

```
FOSGWTool.sh -update_ldap_properties
```

query_ldap_properties

Query the properties of the LDAP servers in a lockbox.

Syntax

```
FOSGWTool.sh --query_ldap_properties
```

NOTE: The FOSGWTool for Windows is `FOSGWTool.bat`.

Example

```
FOSGWTool.sh --query_ldap_properties
```

Single LDAP server example

```
FOSGWTool.sh --query_ldap_properties
```

```
Lockbox contains 1 servers:
Server url: ldap://win12r2-dc.ldaps.local Base dn:DC=ldaps,DC=local
Group name:CN=SIO_GRP_1,DC=ldaps,DC=local

The parameters in format matching to --reset_ldap_properties command are:
--server_url_list 'ldap://win12r2-dc.ldaps.local'
--base_dn_list 'DC=ldaps,DC=local'
--group_name_list 'CN=SIO_GRP_1,DC=ldaps,DC=local'
```

Multiple LDAP servers example

```
FOSGWTool.sh --query_ldap_properties
```

```
Lockbox contains 2 servers:
Server url: ldap://win12r2-dc.ldaps.local  Base dn: DC=ldaps,DC=local   Group name:
CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=local  dnFormat: (empty)        filter: (empty)
Server url: ldap://server.ldap.com         Base dn: dc=ldap,dc=com      Group name:
cn=scaleio_QA,ou=People,dc=ldap,dc=com      dnFormat: CN=<USER>,OU=People,DC=ldap,DC=com
 filter: (&(objectClass=*)(cn=<USER>)(memberOf=<GROUP>))
```

The output corresponds to the following parameter configurations in the `--reset_ldap_properties` command:

```
--server_url_list "ldap://win12r2-dc.ldaps.local;ldap://server.ldap.com"
--base_dn_list "DC=ldaps,DC=local;dc=ldap,dc=com"
--group_name_list
"CN=SIO_GRP_1,OU=SIO_OU_1,DC=ldaps,DC=local;cn=scaleio_QA,ou=People,dc=ldap,dc=com"
--dn_format_list ";CN=<USER>,OU=People,DC=ldap,DC=com" --filter_list ";(&(objectClass=*)
(cn=<USER>)(memberOf=<GROUP>))"
```

7

Configuring LDAP authorization on the LIA

This section explains how to configure the PowerFlex Lightweight Installation Agent (LIA) to use LDAP authentication following PowerFlex deployment. It also contains instructions for adding LDAP servers for LIA LDAP authentication in an existing PowerFlex system.

NOTE: For instructions for configuring LDAP authentication for LIA during PowerFlex installation for a new deployment, see the *Deploy Dell PowerFlex Guide*.

Changing LIA authentication to LDAP

Use the following procedure to change the Lightweight Installation Agent (LIA) authentication method from native to LDAP in a PowerFlex system that was already deployed. LIA is installed on every node during PowerFlex deployment. LIA is used to upgrade the components on the host on which is it installed and is required for many maintenance operations.

LIA supports both native and LDAP authentication for authorization to upgrade a component. If native authentication was specified for LIA during initial PowerFlex Gateway installation, you can switch LIA authentication to LDAP after PowerFlex deployment.

NOTE: After switching LIA authentication to LDAP, assistance from Customer Support is required to change it back to native authentication.

Configure LDAP authentication for LIA on the PowerFlex Gateway

If Linux LDAP servers are used for authenticating LIA, add the Linux LDAP server parameters to the `gatewayUser.properties` file before switching the LIA authentication mode from native to LDAP in an existing PowerFlex system. If a Windows LDAP server is used, skip this task.

Steps

1. Log in to the PowerFlex Gateway server and open the `gatewayUser.properties` file, located at `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes`, for editing.
2. Under `# properties for add lia ldap server`, add the following properties, based on your LDAP server settings.

   - ```
     # properties for add lia ldap server
     lia.ldap.replaceExisting=
     lia.ldap.usernameDnFormat=
     lia.ldap.authorizerDn=
     lia.ldap.searchFilterFormat=
     ```

     - The value for `lia.ldap.replaceExisting=` should be left blank, unless an existing LDAP server configuration on the PowerFlex Gateway is being overwritten.
     - Use the values configured in your organization's LDAP servers for the following parameters:

       ```
       lia.ldap.usernameDnFormat=
       lia.ldap.searchFilterFormat=
       ```

     - The value for the `lia.ldap.authorizerDn=` should be left blank.

     For example:

     ```
     # properties for add lia ldap server
     lia.ldap.replaceExisting=
     lia.ldap.usernameDnFormat=CN=[USER],OU=People,DC=ldap,DC=com
     lia.ldap.authorizerDn=
     lia.ldap.searchFilterFormat=(&(objectClass=userSecurityInformation)(cn=[USER])
     (memberOf=[GROUP]))
     ```
     ```

3. Restart the PowerFlex Gateway service:
   - On a Linux server run:

     ```
     /etc/init.d/scaleio-gateway restart
     ```

   - On a Windows server: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.

Next steps

Continue to the next task to change the authentication method using the PowerFlex Installer.

Change LIA authentication method to LDAP

Use the PowerFlex Installer to switch LIA's authentication to LDAP on an existing PowerFlex system. You can switch LIA's authentication method from native to LDAP, but after you perform this task you will not be able to switch back to native authentication.

Prerequisites

Ensure the following:

- You know the LIA password.
- If a Linux LDAP server is used, you have added the Linux server properties to the `gatewayUser.properties` file, as described in the previous task.

Steps

1. From a web browser, enter the IP address of the PowerFlex Gateway, and log in to the PowerFlex Installer.
2. On the **Maintain** tab, enter the following information:
   a. Primary MDM IP address
   b. MDM admin username and password.
   c. LIA password
3. Click **Retrieve system topology**.
   The PowerFlex Installer displays a list of MDMs and their LIA authentication method.
4. On the **Maintain** tab, click **Security Settings** > **Add LDAP server to LIA authentication list**.
5. In the **Add LDAP server to LIA authentication list** window, enter the MDM admin password, and then fill in the LDAP server URI, group identifier, and BaseDN identifier.

   Example:

   - LDAP Server URI: ldaps://server.ldap.com
   - LDAP Group: cn=scaleio_QA,ou=People,dc=ldap,dc=com
   - LDAPBaseDN: dc=ldap,dc=com
6. Click **Add LDAP server**.
7. Click **Security Settings** > **Change LIAs authentication method to LDAP**.
8. In the **Change LIAs authentication method to LDAP for PowerFlex system** window, enter the MDM admin password, and LDAP username and password.
9. Optionally select the **Force LDAP authentication mode** check box so that LDAP authentication is applied to all accessible LIAs.
10. Click **Change LIAs authentication method to LDAP**.
11. On the **Monitor** tab, confirm that the operation was successful, and then click **Complete**.

Results

LDAP is now set as the LIA authentication method, and credentials for an LDAP server have been added to the PowerFlex Gateway.

Add an additional LDAP server for LIA

PowerFlex supports up to eight LDAP servers for LIA authentication. The first LDAP server's credential are added to the PowerFlex Gateway when LDAP is configured as the LIA authentication method. Use the following procedure to add additional LDAP servers' credentials following PowerFlex deployment.

Prerequisites

Ensure you know the LIA password.

Steps

1. From a web browser, enter the IP address of the PowerFlex Gateway, and log in to the PowerFlex Installer.
2. On the **Maintain** tab, enter the following information:
   a. Primary MDM IP address
   b. MDM admin username and password.
   c. LIA credentials
3. Click **Retrieve system topology**.
4. Click **Security Settings** > **Add LDAP server to LIA authentication list**.
5. In the **Add LDAP server to LIA authentication list** dialog box, enter the MDM admin password, and then fill in the LDAP server URI, group identifier, and BaseDN identifier.

   Example:

   - LDAP Server URI: ldaps://server.ldap.com
   - LDAP Group: cn=scaleio_QA,ou=People,dc=ldap,dc=com
   - LDAPBaseDN: dc=ldap,dc=com
6. Click **Add LDAP Server**.
7. On the **Monitor** tab, confirm that the operation was successful, and then click **Complete**.

Results

The LDAP server's credentials are added to the PowerFlex Gateway for LIA authentication.

Remove an LDAP server from LIA LDAP configuration

Use the following procedure to remove an LDAP server from the PowerFlex Gateway's LIA LDAP configuration. If only one LDAP server is configured for LIA, you must add another server before removing the existing one.

Prerequisites

Ensure you know the LIA credentials.

Steps

1. From a web browser, enter the IP address of the PowerFlex Gateway, and log in to the PowerFlex Installer.
2. On the **Maintain** tab, enter the following information:
   a. Primary MDM IP address
   b. MDM admin username and password.
   c. LIA credentials
3. Click **Retrieve system topology**.
4. On the **Maintain** tab, click **Security Settings** and select **Remove LDAP Server from LIA auuthetication list**
5. In the **Remove LDAP Server from LIA auuthetication list** dialog box, enter the MDM admin password and the URI of the LDAP server to be removed.
6. Click **Remove LDAP Server**.
7. On the **Monitor** tab, confirm that the operation was successful, and then click **Complete**.